INCH-POUND
MIL-DTL-16377/30B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/30A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING FOR PSYCHIATRIC WARDS, 100 WATTS, 120 VOLTS, DRIPPROOF 15 DEGREES, SYMBOLS 137 AND 137SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 7 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

Distribution Statement A. Approved for public release: distribution is unlimited.

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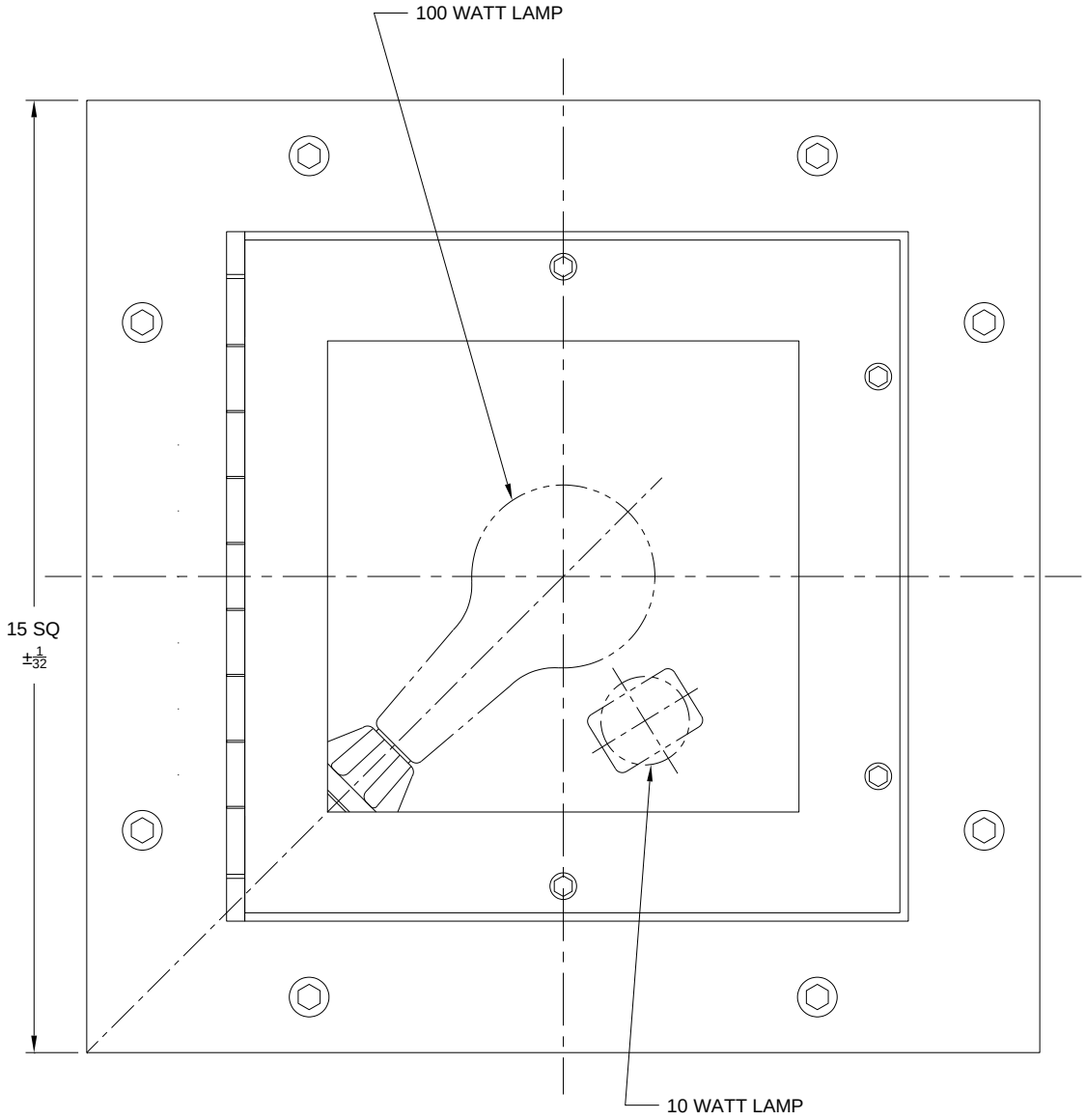


FIGURE 1. Typical view of symbol 137.

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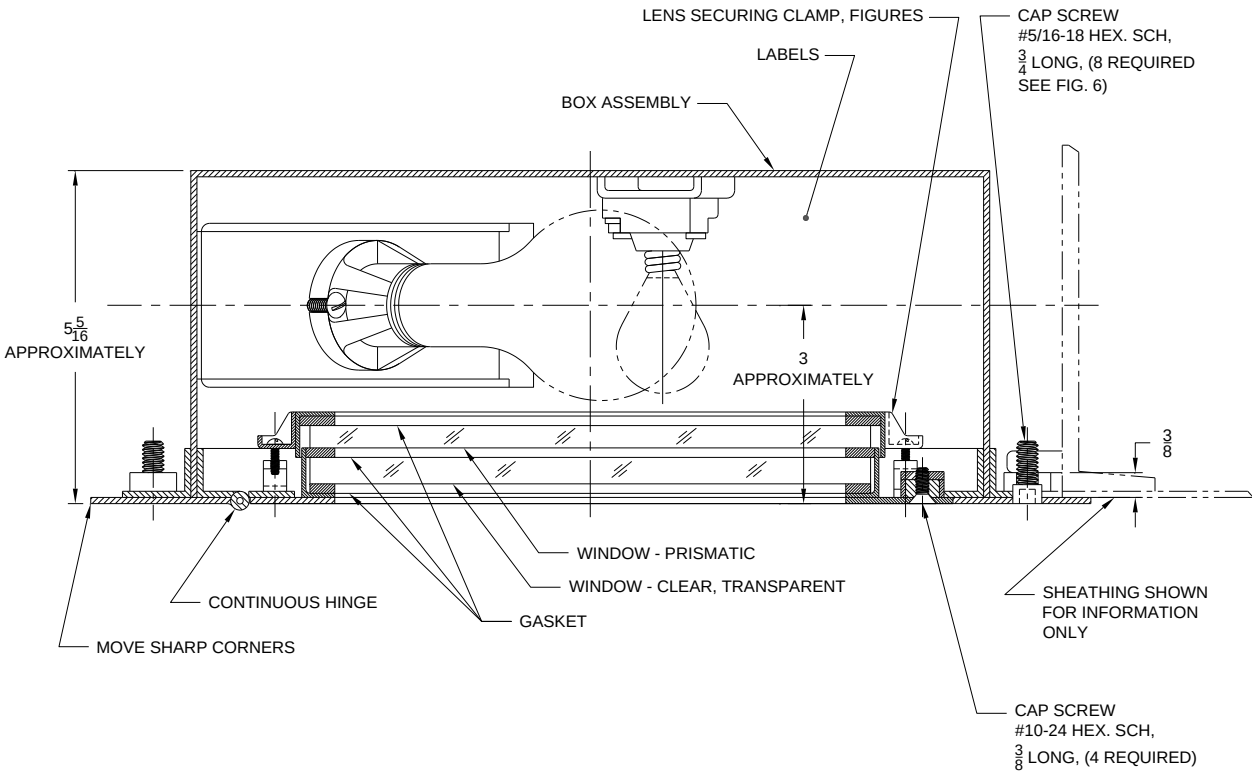


FIGURE 2. Typical section.

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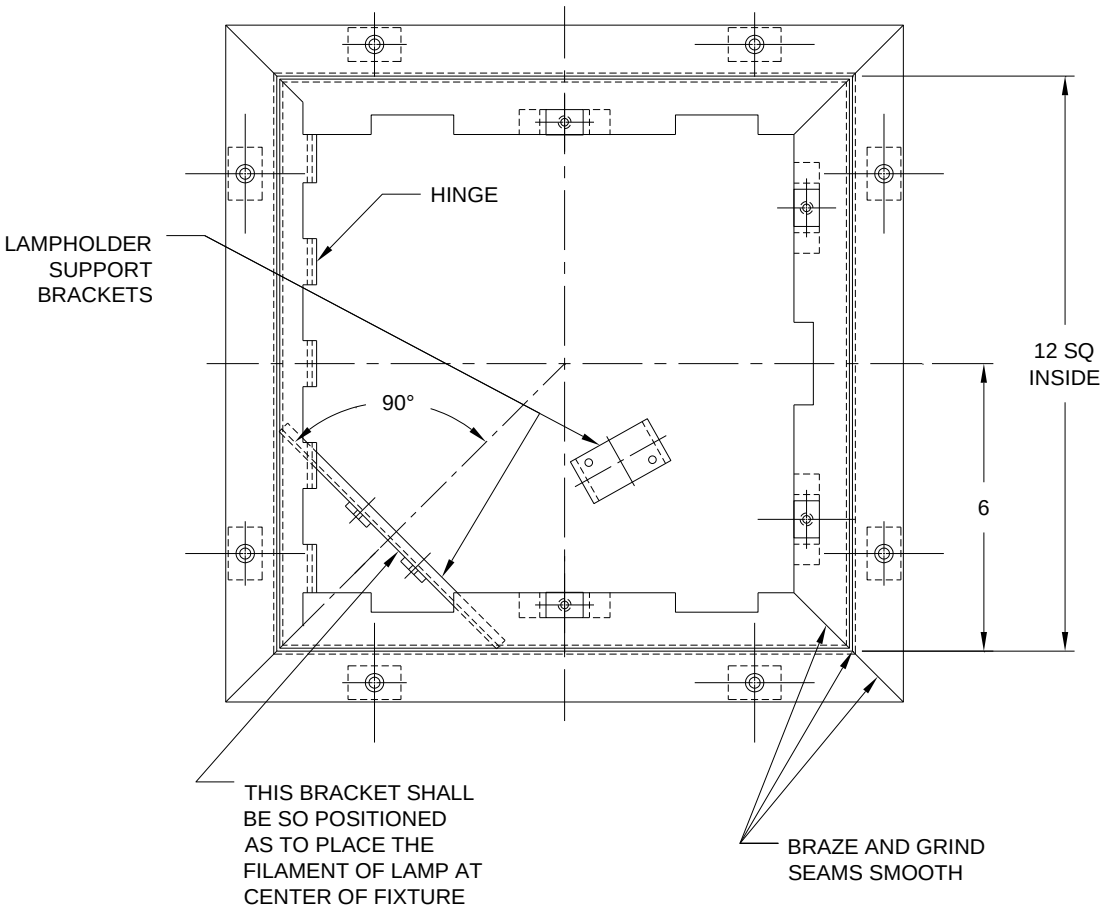


FIGURE 3. Box assembly.

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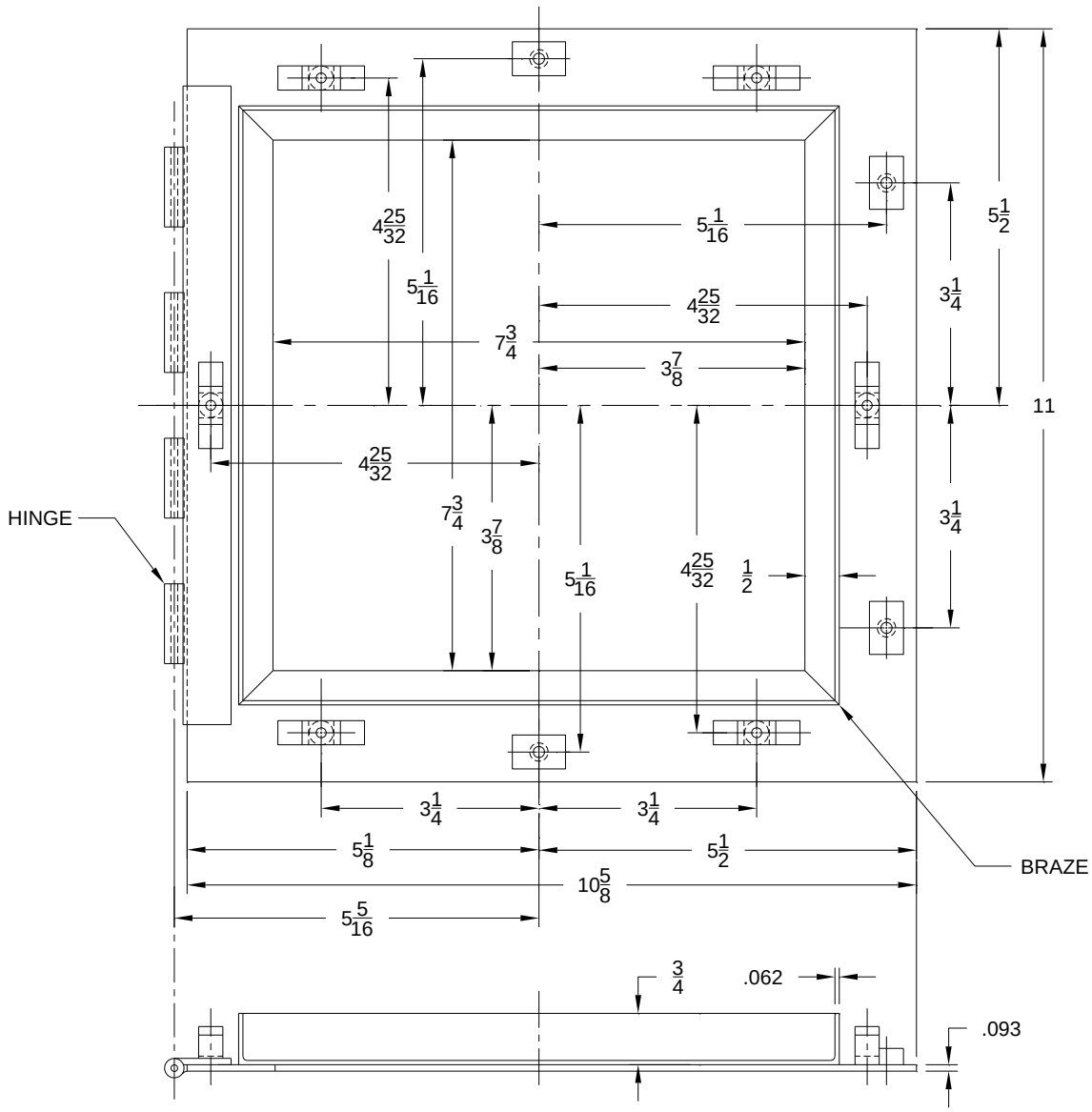


FIGURE 4. Cover assembly.

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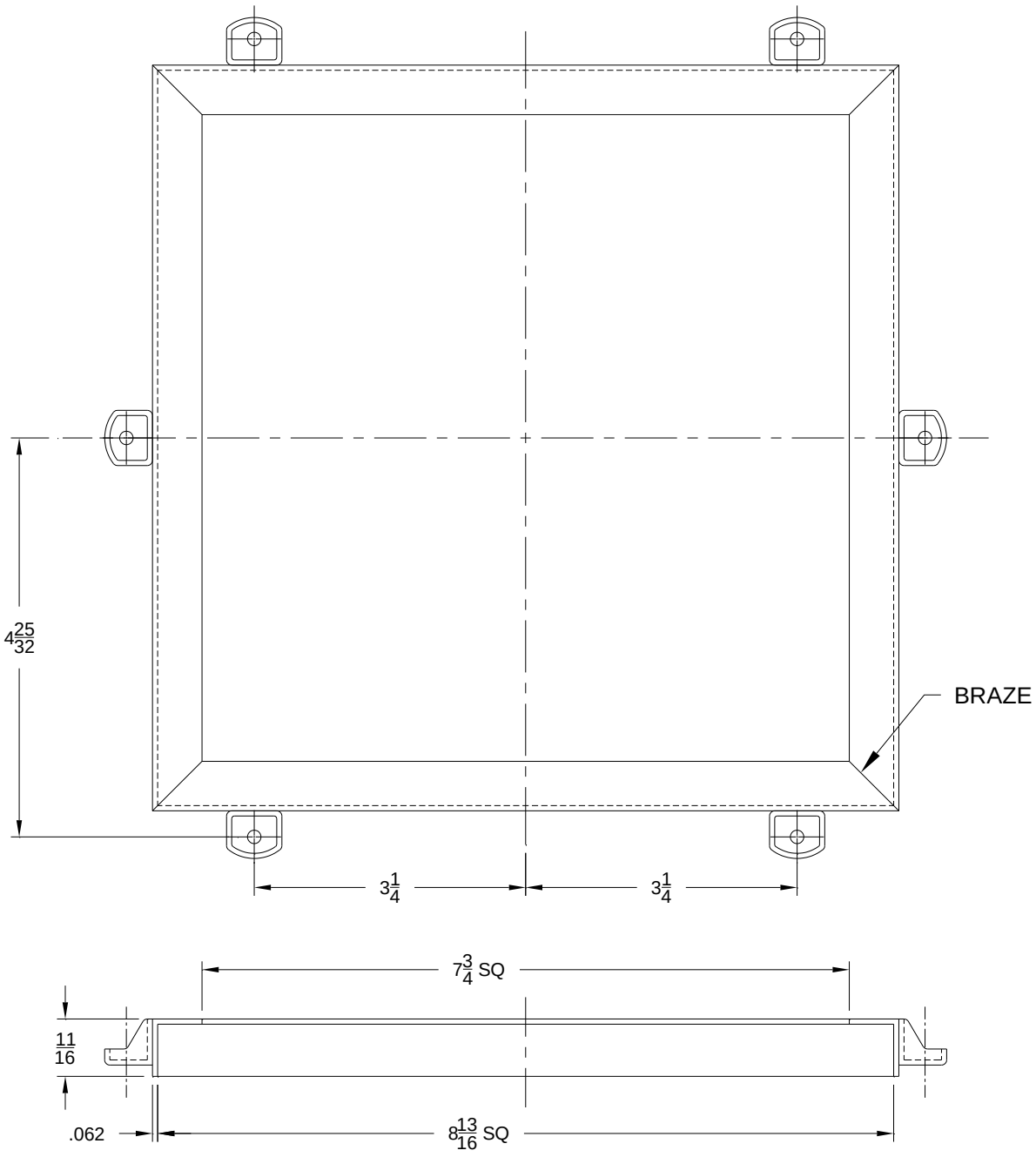


FIGURE 5. Lens securing clamp.

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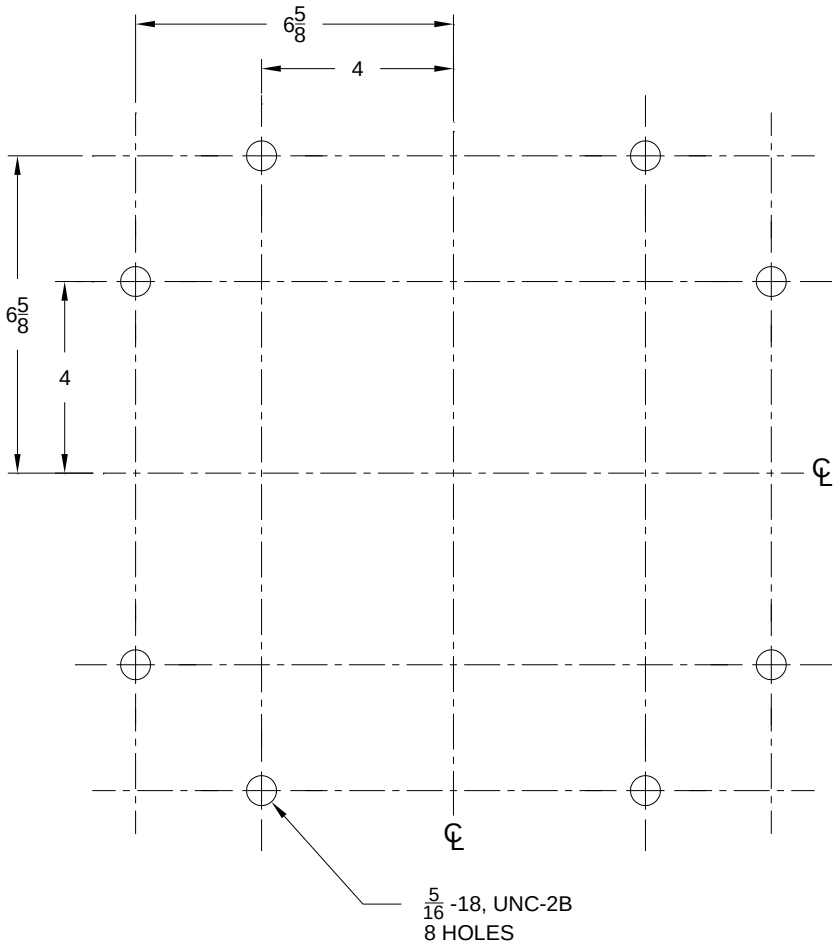
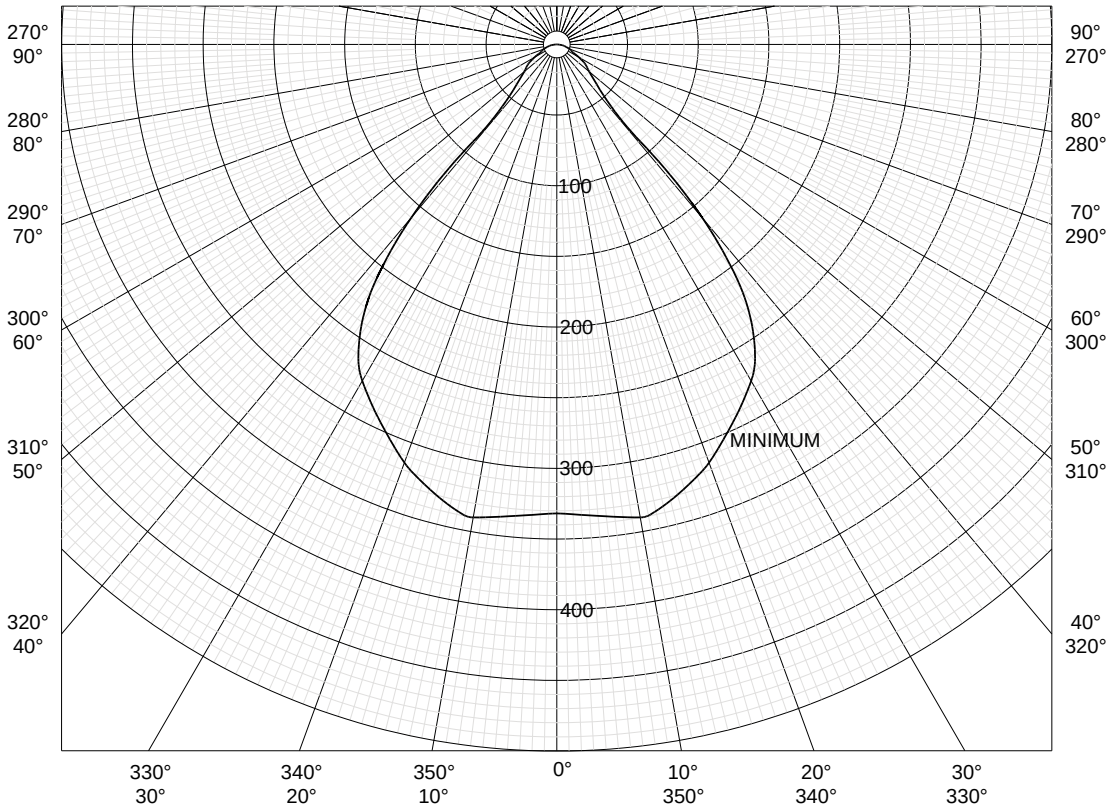


FIGURE 6. Mounting dimensions.

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ANGLE	CANDLEPOWER
	SYMBOL 137
0	332
5	335
10	340
15	330
20	315
25	295
30	275
35	240
40	160
45	40
50	30
55	25
60	18

ANGLE	CANDLEPOWER
	SYMBOL 137
65	12
70	10
75	7
80	5
85	2
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

Photometric test data:

Distance: 10 feet

Voltage: 120 volts

Ambient temperature: 76 °F

Lamps: 100 watts, rough service, conforming to W-L-101/77, will be furnished by laboratory.

FIGURE 7. Candlepower distribution curve.

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REQUIREMENTS

Type: II
Class: 2

SYMBOL NUMBER

137: Fixture for use with one 100-watt white lamp and one 10-watt white lamp for overhead general lighting in psychiatric wards.

DESIGN

Electrical rating:	100 watts, 120 volts.
Dimensions and configuration:	See figures 1 through 6.
Enclosure, cover lens securing clamp, face plates and brackets:	Steel.
Window:	Clear, transparent, heat-resistant, high-impact glass, 8½ by 8½ by ⅜ inches, having an 85-percent-minimum light transmittance.
Window:	Clear, prismatic, Holophane Controlens Catalogue No. 739, or equal.
Lampholders:	
For use with 100-watt lamp:	Medium screw, UL 496, Supplement SB, two contact.
For use with 10-watt lamp:	Candelabra screw, UL 496, Supplement SB.
Hardware:	All exterior screws shall be recessed to provide a smooth, flush surface to the fixture and shall be of the recessed socket type.
Lamp:	This fixture shall be designed to accommodate one 100-watt white lamp in accordance with industry number 100A/RS(120) and one 10-watt lamp in accordance with industry number 10S11/79. (Lamp not furnished.)
Weight:	23 pounds, 7 ounces, maximum.
Labels:	Labels shall be furnished in accordance with MIL-DTL-16377. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be: "INSTALL ONLY: 100-WATT LAMP, NSN 6240-00-246-5052 AND 10-WATT LAMP, NSN6240-00-143-3063"

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBER

137SSL: Fixture for use with two illumination states, one higher illumination state and one lower illumination state, white, for overhead general lighting in psychiatric wards.

DESIGN

Electrical rating:	50 watts maximum, 120 volts, 60 hertz for high output, 10 watts maximum, 120 volts, 60 hertz for low output.
Dimensions and configuration:	See figures 1 through 5.
Photometric data, high output:	See figure 7 of this specification sheet and 3.7.1 of MIL-DTL-16377.
Photometric data, low output:	8 to 12 percent of the high output lumens.

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Wiring:	The high and low outputs shall be separate single-phase circuits that are electrically isolated from each other. The electrical connection devices chosen to terminate incoming power for each circuit (see 3.6.11.1 of MIL-DTL-16377) shall be sized to allow for both an input and output cable so that the fixtures can be installed in series within a space without the use of external junction boxes.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Mounting dimensions:	See figure 6.
Enclosure:	Dripproof. See 3.5.11 of MIL-DTL-16377.
Weight:	23 pounds, 7 ounces, maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 2 and 3.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I for type II and table III for type III.
Windows:	See table II.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table IV.

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TABLE I. First article and comparison inspection for complete type II fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14	Dripproof
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Transmittance	3.7.3.1	4.8.6.3	M16377/30-001 only

TABLE II. First article and comparison inspection for windows.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Transmittance	3.7.3.1	4.8.6.3	M16377/30-001 only

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TABLE III. First article and comparison inspection for complete type III fixtures.[1/]

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	[1/]
Insulation resistance	3.6.16	4.8.3	[1/]
Enclosure effectiveness	3.5.11	4.8.14.4	[1/] Dripproof
Shock	3.5.12	4.8.7	[1/] Grade A
Vibration	3.5.13	4.8.8	[1/]
Enclosure effectiveness	3.5.11	4.8.14.4	[1/] Dripproof
Dielectric withstanding voltage	3.6.15	4.8.2	[1/]
Insulation resistance	3.6.16	4.8.3	[1/]
Noise	3.5.14	4.8.9	[1/]
Continuity of grounding	3.6.13.4	4.8.12	[1/]
Photometrics	3.7.1	4.8.6	---
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

[1/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. Part umber and general information.

Military Part Number M16377/30-	Item Name	National Stock Number (NSN)	Supersedes Part Number
137	Complete fixture, symbol 137	6210-00-152-2844	MS17316-137
100	Lens, clear transparent	---	9000-S6401-74222-96
002	Lens, clear prismatic	---	9000-S6401-74222-97
137SSL	Complete fixture, symbol 137SSL	---	---

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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